

A LINEAR PROGRAMMING CAPSTONE

Cultivating Caraga

Optimizing Agricultural Land Allocation Across Five Provinces

Data Analyst Team | Region XIII — Caraga



WHY AGRICULTURE?

The smallest slice of the pie — and the most contested

SERVICES

57.0%

PhP 218.66B • 2025 GRDP

INDUSTRY

32.9%

PhP 126.37B • 2025 GRDP

AGRICULTURE, FORESTRY & FISHERY

10.1%

PhP 38.69B • 2025 GRDP

Caraga grew 5.7% in 2025 — 2nd fastest region in the country. Agriculture is the smallest slice of that growth — but it's also the sector where land, food security, and farmer livelihoods collide most directly.

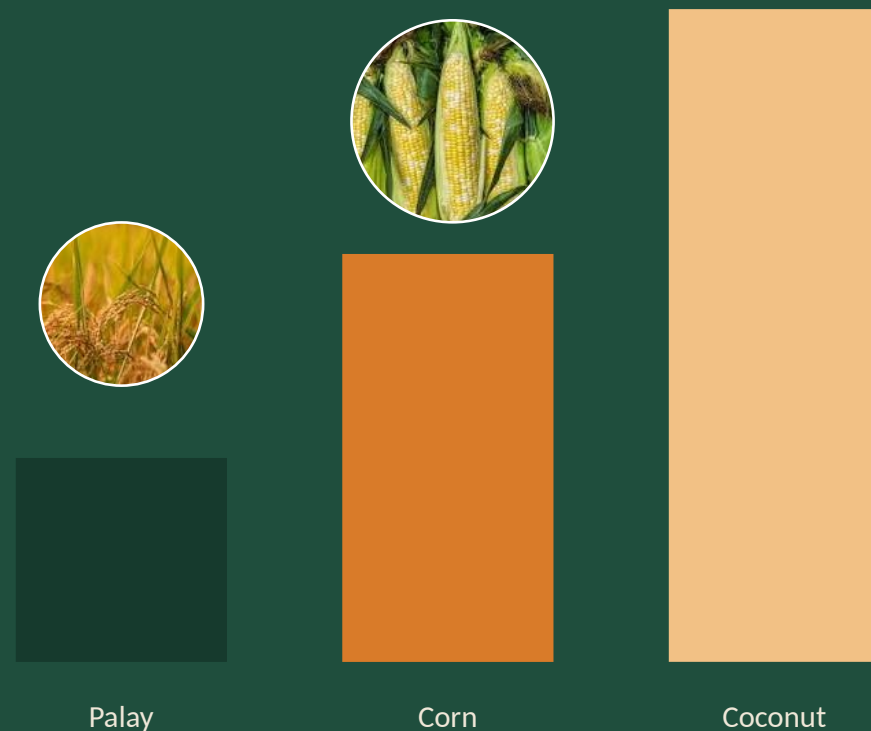
Not every profitable choice is a good one

Coconut earns roughly 3x more profit per hectare than palay.

A model that only chases profit would push every province toward 100% coconut — and away from the rice and corn that feed the region.

The real question: "How do we make the most money from farming, without accidentally stopping people from growing enough food to eat?"

Illustrative profit/ha comparison



WHY THESE THREE CROPS

They’re not our pick — they’re Caraga’s

Share of Caraga's major-crop land area, 2023

Crop	Share
Coconut	43.7%
Palay	36.4%
Corn	8.3%
Banana	5.3%
Rubber	2.9%
Coffee	1.1%
Others (cacao, mango, cassava, soybean, veg.)	~1.6%

PALAY + CORN + COCONUT

~88%

of all major-crop land in Caraga

Everything else combined — banana, rubber, coffee, cacao, and more — is a rounding error by comparison.

Source: DA-Caraga Annual Report 2023 (area by crop); PCA Coconut Statistics 2023 (coconut bearing area)

OUR MISSION

How should Caraga allocate its farmland across palay, corn, and coconut — across five provinces — to maximize profit, while making sure rice and corn farming don't disappear?

THE CENTRAL QUESTION

The Work Pipeline — Five Acts

01

Collect

Pull real production, area, and cost data from PSA and PCA across all five provinces.

02

Calculate

Turn raw numbers into yield/ha and net returns/ha — the LP's actual coefficients.

03

Build

Define 15 decision variables and the profit-maximizing objective function.

04

Constrain

Add land limits, food-security floors, and realistic crop-share caps per province.

05

Solve & Present

Solve the model, sanity-check the result, and tell the allocation story.

Note: an air-quality (AQI) scenario layer was explored and is scoped as future work — see closing slide.

STEP 01 — COLLECT

The Raw Numbers We Start With

Note: 1 metric tons = 1000kg

Production and area harvested by province — most recent confirmed year per crop

Province	Palay Produce (Metric Tons)	Palay (hectares)	Corn Produce (MT)	Corn (hectares)	Coconut Produce (MT)	Coconut (hectares)
Agusan del Norte	83,876	24,388	19,119	5,137	163,805.50	34,500
Agusan del Sur	312,379.60	89,848	102,353.20	29,690	45,165.97	14,888
Surigao del Norte	69,701	19,195	2,287	880	52,451.69	60,122.92
Surigao del Sur	101,092	28,606	14,512	5,055	377,693.75	81,175.84
Dinagat Islands	7,061	2,852	347	172	3,822.76	4,572.49
CARAGA TOTAL	574,092.72	164,900	138,614.40	40,933	642,939.67	195,259.25

Corn area harvested shown for 2021 — 2024 corn area by province was not available (flagged data gap).

Sources: PSA Region XIII Palay & Corn Production Surveys (2021, 2024); Philippine Coconut Authority, Coconut Statistics (2023)

STEP 02 — CALCULATE

Turning Raw Numbers into Model Coefficients

YIELD PER HECTARE

Yield (MT/ha) = Production (MT) ÷ Area Harvested (ha)

Example — Caraga Coconut, 2023:

642,939.67 MT ÷ 195,259.25 ha = **3.29 MT/ha**

NET RETURNS PER HECTARE (PROFIT)

Net Returns = Gross Returns – Total Costs

Example — Caraga Palay, 2022:

₱53,383 – ₱42,610 = **₱10,773 / ha**

Production (kg/hectare) = Yield × (1000)

GROSS RETURNS → FEEDS DIRECTLY INTO NET RETURNS ABOVE

Gross Returns = Production (kg per hectare) × Farmgate Price (PhP/kg) e.g. Coconut: 3,290 kg/ha × ₱11.43/kg = ₱37,604.70/ha

These per-hectare figures — not the raw PSA tables — are what plug directly into the LP's objective function.

HONESTY CHECK

The Data We Stand On

We didn't hide the gaps — we built the model around them.

Data Category	Coverage	Confidence
Palay production & area - from year 2024	5 provinces × 3 years	HIGH — PSA direct
Corn production (2024) & area (2021)	5 provinces × 2–3 years	HIGH — PSA direct
Coconut production & area - from year (2023	5 provinces × 1 year	MEDIUM — PCA snapshot
Palay costs & returns - from year 2022	Caraga-specific, 2022	HIGH — PSA direct
Corn costs & returns - from year 2022	National proxy	MEDIUM — flagged proxy(national figure)
Coconut costs & returns - 2023 & 2024	Stitched estimate	LOW — flagged estimate
Air quality (PM2.5)	Descoped this phase	FUTURE WORK

Each province, a different crop story

THE RICE BASKET

Agusan del Sur

54%+

of regional palay production

THE COCONUT GIANT

Surigao del Sur

59%

of regional coconut production

MOST DIVERSIFIED

Agusan del Norte

3 crops

all significantly present on its land

STEADY PRODUCER

Surigao del Norte

3rd

largest palay area in the region

THE SMALLEST FOOTPRINT

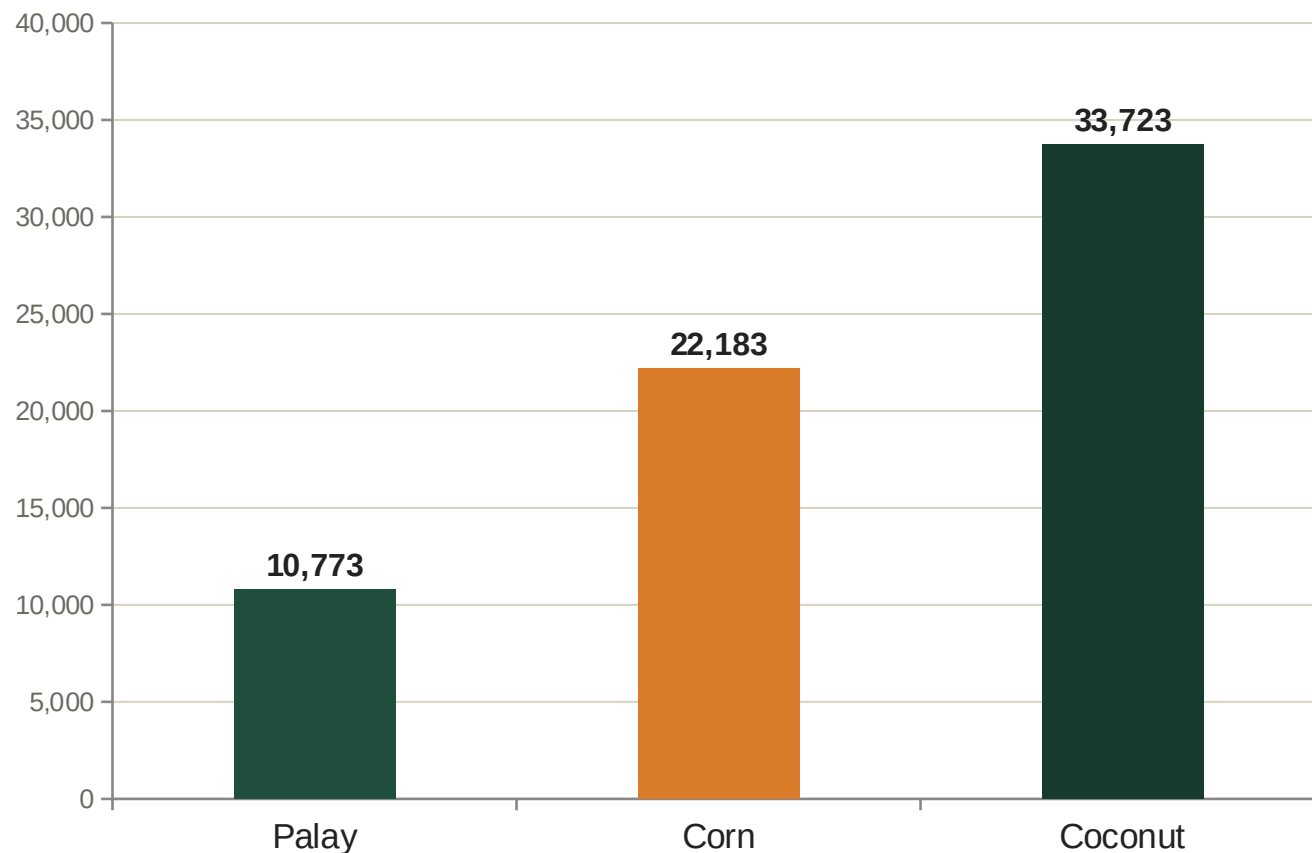
Dinagat Islands

~1%

share of regional crop production

STEP 02 RESULT

The Profit Picture (per hectare)



Net Returns = Gross Returns – Total Costs

Explain harvest season

The tension the model has to resolve:

If coconut is this profitable, why isn't all farmland already coconut?

Tree maturity, land history, and food security are real-world limits the LP has to respect — not override.

 *Coconut figure is a stitched, lower-confidence estimate — see Data We Stand On.*

Our First Run Broke the Model — On Purpose, We Fixed It

ATTEMPT 1 — PURE PROFIT MAXIMIZATION

100% Coconut

Every province, every hectare — the model abandoned palay and corn entirely.

Why it happened: with no limits, an LP always pushes every hectare to whichever crop has the single highest profit coefficient.

THE FIX — REALISTIC CONSTRAINTS

Add Real-World Floors & Caps

- Palay & corn floors — 80% of current planted area, so food crops can't be abandoned
- Coconut caps — current share + a realistic growth buffer per province
- Land ceilings — based on hectares already in use for these 3 crops

Meet the Decision Variables

Every decision the model can make is: how many hectares of Crop J go to Province I?

	Palay	Corn	Coconut
Agusan del Norte	x_1	x_2	x_3
Agusan del Sur	x_4	x_5	x_6
Surigao del Norte	x_7	x_8	x_9
Surigao del Sur	x_{10}	x_{11}	x_{12}
Dinagat Islands	x_{13}	x_{14}	x_{15}

x_i = hectares allocated to a specific crop, in a specific province. **15 variables total (5 provinces × 3 crops).**

The Objective Function

Maximize total profit for the caraga across all 15 variables

Maximize $Z = 10,773 \times (\text{Palay hectares, all provinces}) + 22,183 \times (\text{Corn hectares, all provinces}) + 33,723 \times (\text{Coconut hectares, all provinces})$

or

Maximize $Z = 10,773x_1 + 22,183x_2 + 33,723x_3 + 10,773x_4 + 22,183x_5 + 33,723x_6 + 10,773x_7 + 22,183x_8 + 33,723x_9 + 10,773x_{10} + 22,183x_{11} + 33,723x_{12} + 10,773x_{13} + 22,183x_{14} + 33,723x_{15}$

Palay

₱10,773 / ha

Caraga-specific net returns

Corn

₱22,183 / ha

National proxy, flagged

Coconut

₱33,723 / ha

Stitched estimate, flagged

The Constraints

21 constraints keep the model realistic: land limits, food-security floors, and growth caps

Province	Land Cap (ha)	Palay Min (80%)	Corn Min (80%)	Coconut Max
Agusan del Norte	64,025.00	19,510.40	4,109.60	44,104
Agusan del Sur	134,426.00	71,878.40	23,752.00	35,052
Surigao del Norte	80,197.92	15,356.00	704.00	72,153
Surigao del Sur	114,836.84	22,884.80	4,044.00	98,401
Dinagat Islands	7,596.49	2,281.60	137.60	5,711

21 constraints total: 5 land caps + 5 palay floors + 5 corn floors + 5 coconut caps + 1 non-negativity (all $x_i \geq 0$)

Palay/corn floors set at 80% of current planted area — protects food security while still allowing reallocation.

The Constraints, Written Out in Full

LAND CONSTRAINTS (5)

$$x1 + x2 + x3 \leq 64,025.00 \quad (\text{ADN})$$

$$x4 + x5 + x6 \leq 134,426.00 \quad (\text{ADS})$$

$$x7 + x8 + x9 \leq 80,197.92 \quad (\text{SDN})$$

$$x10 + x11 + x12 \leq 114,836.84 \quad (\text{SDS})$$

$$x13 + x14 + x15 \leq 7,596.49 \quad (\text{DI})$$

PALAY MINIMUM — 80% FLOOR (5)

$$x1 \geq 19,510.40 \quad x4 \geq 71,878.40$$

$$x7 \geq 15,356.00 \quad x10 \geq 22,884.80$$

$$x13 \geq 2,281.60$$

NON-NEGATIVITY (1)

$$x1, x2, x3, \dots, x15 \geq 0$$

CORN MINIMUM — 80% FLOOR (5)

$$x2 \geq 4,109.60 \quad x5 \geq 23,752.00$$

$$x8 \geq 704.00 \quad x11 \geq 4,044.00$$

$$x14 \geq 137.60$$

COCONUT MAXIMUM (5)

$$x3 \leq 44,104 \quad x6 \leq 35,052$$

$$x9 \leq 72,153 \quad x12 \leq 98,401$$

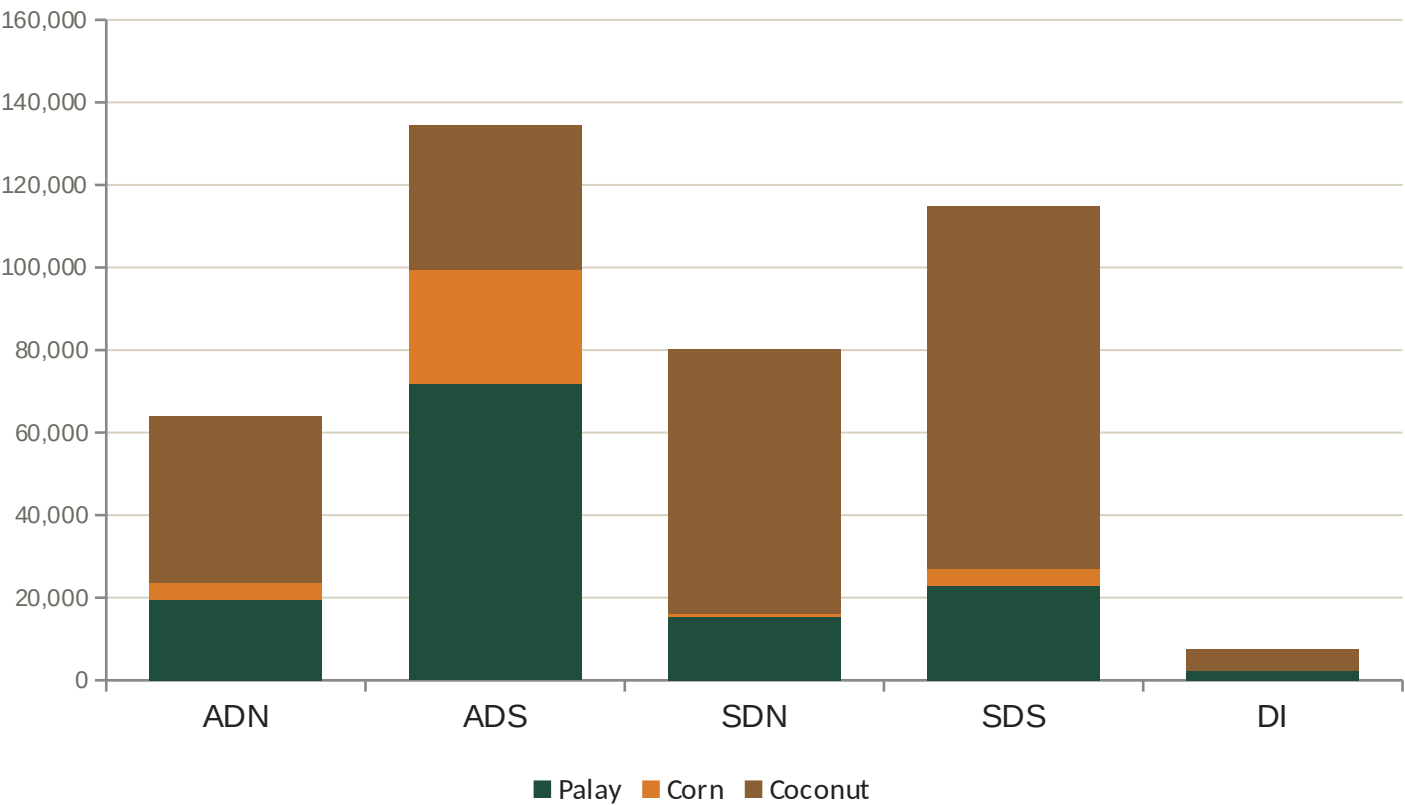
$$x15 \leq 5,711$$

$$\text{Ceiling} = \text{Curent Land} + (15\% \times \text{Total Land})$$

$$\text{Floor} = \text{Current Land} (80\%)$$

The Optimal Allocation

Solved with PuLP (CBC solver) — Status: Optimal



MAXIMUM TOTAL PROFIT (Z)

₱10.08 Billion

across all five provinces, per cropping cycle

What the solution shows:

- Palay & corn sit exactly at their 80% food-security floor in every province.
- Coconut absorbs all remaining land — maxed out completely in Agusan del Sur.
- All available farmland is fully utilized in every province.

Solution lands on constraint boundaries (a “corner point”) — expected LP behavior, not an error.

The Numbers Behind the Chart

Solved allocation, hectares per province and crop

Province	Palay (ha)	Corn (ha)	Coconut (ha)	Total (ha)
Agusan del Norte	x1 = 19,510.40	x2 = 4,109.60	x3 = 40,405.00	64,025.00
Agusan del Sur	x4 = 71,878.40	x5 = 27,495.60	x6 = 35,052.00	134,426.00
Surigao del Norte	x7 = 15,356.00	x8 = 704.00	x9 = 64,137.92	80,197.92
Surigao del Sur	x10 = 22,884.80	x11 = 4,044.00	x12 = 87,908.04	114,836.84
Dinagat Islands	x13 = 2,281.60	x14 = 137.60	x15 = 5,177.29	7,596.49

Status: Optimal Maximum Total Profit (Z) = ₱10,077,230,844.75 (≈ ₱10.08 Billion)

Solved using PuLP (CBC solver) in Python/Jupyter Notebook.

A profit-maximizing, foodland-security-conscious allocation for Caraga

SCOPED FOR FUTURE WORK

An AQI/PM2.5 scenario layer for Agusan del Norte — exploring how worsening air quality could shift the optimal allocation — was investigated during this project. It's a natural next phase once province-level air-quality monitoring expands beyond the current single station.

Thank you.

QUESTIONS YOU MIGHT ASK

Q1: Why does Corn have a higher profit per hectare than Palay?

Palay is more input-intensive to grow — it requires irrigation, more hired labor, and higher overall production costs (₱42,610/ha vs. Corn's ₱34,710/ha) — even though both crops earn similar gross revenue per hectare. Corn simply costs less to produce, so more of its revenue stays as profit.

Q2: If Coconut is the most profitable crop, why doesn't the model just plant 100% coconut everywhere?

Because we added real-world constraints. Coconut trees take years to mature, and rice/corn are food-security crops that can't be abandoned. We capped coconut's growth and set an 80% minimum floor for palay and corn in every province, forcing the model to stay realistic instead of chasing pure profit.

QUESTIONS YOU MIGHT ASK

Q3: What would i do differently with more time or better data?

Get province-level, single-year cost/return data for corn and coconut (not proxies), fill the 2022/2023/2025 production gaps, and properly reintroduce the AQI scenario using a yield-based (not land-allocation-based) model structure.

Q4. Why choose the 80% as floor and 15% as ceiling

To make sure the model can actually actually reallocate/move land without neglecting the opportunity within coconut and won't neglect the other crops.